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Case 32-2020: A 63-Year-Old Man with Confusion, Fatigue, and Garbled Speech

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PRESENTATION OF CASE

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Dr. Kevin L. Ard: A 63-year-old man was evaluated at this hospital in the spring because of fever, confusion, headache, garbled speech, fatigue, vision changes, lymphocytic pleocytosis, and an elevated protein level in the cerebrospinal fluid (CSF).

Approximately 5 years before the current presentation, the patient had received a diagnosis of primary membranous nephropathy at another hospital. Treatment with oral prednisone and cyclosporine was initiated.

Two years later, the patient was referred to the nephrology clinic of this hospital for worsening proteinuria and hypertension; rituximab, prednisone, and cyclophosphamide were administered. During the next 21 months, he received four doses of rituximab, the last of which was administered 3 months before this presentation.

Three months before this presentation, the patient had a flash of light accompanied by transient sharp pain in the left retro-orbital area and forehead, with monocular blurry vision. He was subsequently evaluated at the other hospital. Magnetic resonance imaging (MRI) and computed tomography (CT) of the head were normal. Intravenous normal saline was administered, and he was discharged. One week later, the patient was reevaluated at the same hospital for an episode of garbled and nonsensical speech with impaired comprehension that resolved after 10 minutes. A CT angiogram and an MRI of the head reportedly showed a subacute infarct in the right side of the pons; intravenous alteplase was administered. Transthoracic echocardiography (TTE) was normal. Mild word-finding difficulties and confusion persisted.

After discharge, the patient had worsening vision with floaters in the left eye, headache, photophobia, sonophobia, and fatigue. Three weeks later, he was evaluated by a neuro-ophthalmologist, who prescribed oral butalbital-acetaminophen-caffeine therapy and ophthalmic prednisolone drops for suspected uveitis. The vision improved, and headaches decreased during the subsequent 2 weeks; how-

Table 1. Laboratory Data.*

Variable	Reference Range, Adults, This Hospital†	1 Wk before Presentation, Other Hospital	On Initial Consultation, This Hospital
Hemoglobin (g/dl)	13.5–17.5	10.8	10.4
Hematocrit (%)	46.0–56.0	31.6	30.2
Red-cell count (per μ l)	4,500,000–5,900,000	3,580,000	3,460,000
White-cell count (per μ l)	4500–11,000	7300	13,970
Differential count (per μ l)			
Neutrophils	1800–7700	5767	11,651
Lymphocytes	1000–4800	876	1173
Monocytes	200–1200	511	1048
Eosinophils	0–900	0	14
Basophils	0–300	0	14
Platelet count (per μ l)	150,000–400,000	269,000	462,000
Sodium (mmol/liter)	135–145	132	133
Potassium (mmol/liter)	3.4–5.0	4.2	4.2
Chloride (mmol/liter)	100–108	95	90
Carbon dioxide (mmol/liter)	23–32	31	29
Urea nitrogen (mg/dl)	8–25	19	19
Creatinine (mg/dl)	0.60–1.50	0.76	0.87
Glucose (mg/dl)	70–110	134	105
C-reactive protein (mg/liter)	<8	39.7	6.7
Erythrocyte sedimentation rate (mm/hr)	0–13	12	8

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for glucose to millimoles per liter, multiply by 0.05551.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

ever, 1 week later, he was readmitted to the other hospital with sudden onset of weakness on the right side, facial droop, and garbled speech. The weakness resolved after 10 minutes. CT and MRI of the head and TTE were reportedly unchanged from the previous imaging studies. An electroencephalogram (EEG) showed focal left temporal slowing, and treatment with levetiracetam was initiated. The garbled speech resolved after 2 days, and he was discharged from the hospital.

During the next week, the patient had recurrent staring spells that lasted for 1 minute, during which his family thought he appeared “vacant”; low-grade fever also developed, and within 48 hours, the temperature increased to 39.4°C and the garbled speech recurred. He was readmitted to the other hospital, and CT of the head was

again reportedly normal. Blood levels of calcium, albumin, globulin, ammonia, thyrotropin, lactate, and troponin T were normal, as were the results of liver-function tests and urinalysis. Tests for Lyme disease antibodies, hepatitis B virus surface antigen, hepatitis C virus antibodies, rapid plasma reagin, antinuclear antibodies, rheumatoid factor, and antineutrophil cytoplasmic antibodies (ANCA) were negative, and levels of complement were normal. Other laboratory test results are shown in Table 1. A lumbar puncture was performed; tests for antibodies to *Borrelia burgdorferi* and West Nile virus were negative, and a Venereal Disease Research Laboratory (VDRL) test of the CSF was nonreactive (Table 2). The patient was transferred to this hospital for further evaluation.

A review of systems provided by the patient’s

Table 2. Cerebrospinal Fluid Data.*

Variable	Reference Range, Adults, Other Hospital	3 Days before Presentation, Other Hospital	Reference Range, Adults, This Hospital†	Hospital Day 2, This Hospital	Hospital Day 5, This Hospital
Opening pressure (cm of water)				10	
Color				Yellow	Colorless
Turbidity				Slightly turbid	Clear
Glucose (mg/dl)	40–70	39	50–75	39	55
Total protein (mg/dl)	15–45	358.0	5–55	415	313
Red-cell count (per μ l)			0	18	3
White-cell count (per dl)		238			
White-cell count (per μ l)			0–10	146	49
Differential count (%)					
Neutrophils				50	
Lymphocytes		88	0–100	25	92
Monocytes			0–100	25	8
<i>Borrelia burgdorferi</i> antibodies		Not detected			
VDRL test		Nonreactive			
West Nile virus antibodies		Not detected			
Additional findings		Gram's staining: abundant neutrophils, mononuclear cells without organisms			
					Cytologic assessment: abundant neutrophils, lymphocytes, and monocytes, including reactive forms
					Flow cytometry: no evidence of monoclonal B-cell or unusual T-cell populations

* VDRL denotes Venereal Disease Research Laboratory.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

family was notable for somnolence, fatigue, generalized weakness, unsteadiness, headaches, staring spells, floaters in the left eye, photophobia, sonophobia, mild neck stiffness, unintentional weight loss of 10 kg in the past 6 months, and nocturia. The patient reported no muscle or joint pain, rash, eye pain, diplopia, oral lesions, cough, dyspnea, nausea, or abdominal pain. His medical history was notable for hypertension, dyslipidemia, early-onset coronary artery disease for which he had received percutaneous coronary intervention in his 50s and for which he had undergone aortocoronary bypass grafting in his 60s, congenital color blindness, colonic polyps, and prostatic hypertrophy. Medications included cyclophosphamide, levetiracetam, aspirin, pravastatin, metoprolol, losartan, hydrochlorothiazide, trimethoprim–sulfamethoxazole, omeprazole, calcium carbonate, cholecalciferol, ferrous sulfate, a multivitamin, and fluticasone. He had not taken other antibiotic agents or non-steroidal antiinflammatory drugs.

The patient was a retired government employee and lived with his wife in northern New England. He had recently traveled to Florida and to rural Canada and had spent time hunting. He had smoked one pack of cigarettes per day for approximately 30 years but had quit 7 years before the current presentation. He drank two beers per week and did not use illicit substances. His family history included coronary artery disease in both parents and valvular disease in his mother, psoriatic arthritis in a sister, and sarcoidosis in several siblings; there was no history of renal or neurologic disease.

The temperature was 37.2°C, the heart rate 73 beats per minute, the blood pressure 128/78 mm Hg, and the oxygen saturation 96% while the patient was breathing ambient air. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 20.4. He appeared cachectic, with bitemporal wasting. He was somnolent and difficult to arouse with voice; he was oriented to self and intermittently to place, month, and year. His thoughts were perseverative. On mental-status examination, he was unable to name days of the week backward (he named a month and then named the days forward) and was unable to perform serial sevens calculations. Occasional word-finding difficulties were noted, although repetition, naming, and reading were generally intact. When asked to

remember three words, he could recall only one word after 5 minutes. He reported that nine quarters equaled \$4.25. His affect and speech were intact. The cranial-nerve examination revealed blurriness in the visual fields, "discomfort" with eye movements, and limited upward gaze. There was no weakness or pronator drift; a low-amplitude intention tremor was present in the left hand. The muscle tone was increased in both legs, with diffuse hyperreflexia, pectoralis reflexes, and seven beats of ankle clonus. The remainder of the examination was normal.

Blood levels of calcium, magnesium, phosphorus, glycated hemoglobin, bilirubin, albumin, prealbumin, total protein, lactate dehydrogenase, high-density and low-density lipoprotein cholesterol, and thyrotropin were normal, as were the prothrombin time, partial-thromboplastin time, and results of liver-function tests; other laboratory test results are shown in Table 1. A comprehensive urine toxicology screen was negative, and a serum toxicology screen was negative for acetaminophen, salicylates, ethanol, and tricyclic antidepressants. Screening tests for human immunodeficiency virus (HIV) types 1 and 2 and *Treponema pallidum* were negative. Cultures of the blood and urine were obtained.

Dr. Pamela W. Schaefer: A chest radiograph was normal. CT of the head, performed without the administration of intravenous contrast material, revealed no hemorrhage, acute infarction, or mass lesion. Opacification of the right maxillary sinus was noted. MRI of the head (Fig. 1) and magnetic resonance angiography of the head and neck revealed signal hyperintensity in the sulci on fluid-attenuated inversion recovery images, with associated subtle enhancement on gadolinium-enhanced T1-weighted images; additional subarachnoid and subdural foci characterized by restricted diffusion with no enhancement; non-specific changes in the white matter; opacification of the right maxillary sinus; and no vascular abnormality. MRI of the spine revealed multi-level degenerative changes, without mass effect on the spinal cord, cord signal abnormality, or abnormal enhancement.

Dr. Ard: Intravenous acyclovir, ceftriaxone, ampicillin, vancomycin, and thiamine were administered. On the second hospital day, the temperature was 37.6°C, and myoclonic jerks were noted; marked lethargy, photophobia, and nonsensical speech persisted. The patient was intermittently

impulsive and uncooperative. The white-cell count was 17,960 per microliter (reference range, 4500 to 11,000), and the blood levels of vitamin B₁₂, angiotensin-converting enzyme, complement, and haptoglobin and CD4+ T-cell subset counts were normal. ANCA, antinuclear antibodies, cryptococcal antigen, cytomegalovirus (CMV) DNA, Epstein-Barr virus DNA, galactomannan, and *B. burgdorferi* antibodies were not detected in the blood. A lumbar puncture was performed; the opening pressure was 10 cm of water, and the CSF was yellow and slightly turbid. Additional results of CSF analysis are shown in Table 2. CSF was obtained for culture, cytologic examination, and an autoimmune encephalopathy panel. An EEG showed intermittent periods of theta slowing in the bilateral posterior quadrants, with no epileptiform discharges.

On examination by an ophthalmologist, bilateral visual acuity, visual fields, and intraocular pressure were normal, and there was no afferent pupillary defect. Slit-lamp examination of the eyes revealed bilateral posterior vitreal detachment but no cells in the anterior or posterior chambers. Dilated-eye examination revealed edema in both optic nerves, with tortuous vessels. There was no evidence of anterior uveitis, retinitis, or retinal necrosis.

On the third hospital day, the temperature was 36.7°C, and the white-cell count was 8300 per microliter. The patient reported improved strength. Tests for bartonella antibodies and *Tropheryma whippelii* DNA were negative. Serum protein electrophoresis revealed a diffuse decrease in gamma globulin, with an IgG level of 378 mg per deciliter (reference range, 614 to 1295), an IgA level of 30 mg per deciliter (reference range, 69 to 309), and an IgM level of 7 mg per deciliter (reference range, 53 to 334); the serum free light chain levels and the results of immunoelectrophoresis were normal. A CSF VDRL test was nonreactive, and CSF testing was negative for cryptococcal antigen, eastern equine encephalitis antibodies, enterovirus RNA, West Nile virus RNA, herpes simplex virus (HSV) types 1 and 2 DNA, and varicella-zoster virus (VZV) DNA.

On the fourth hospital day, the patient reported feeling better and having increased strength, which allowed him to walk. Blood tests for cryoglobulin, 1,3- β -D-glucan, and thyroid peroxidase antibodies were negative.

On the fifth hospital day, the patient was

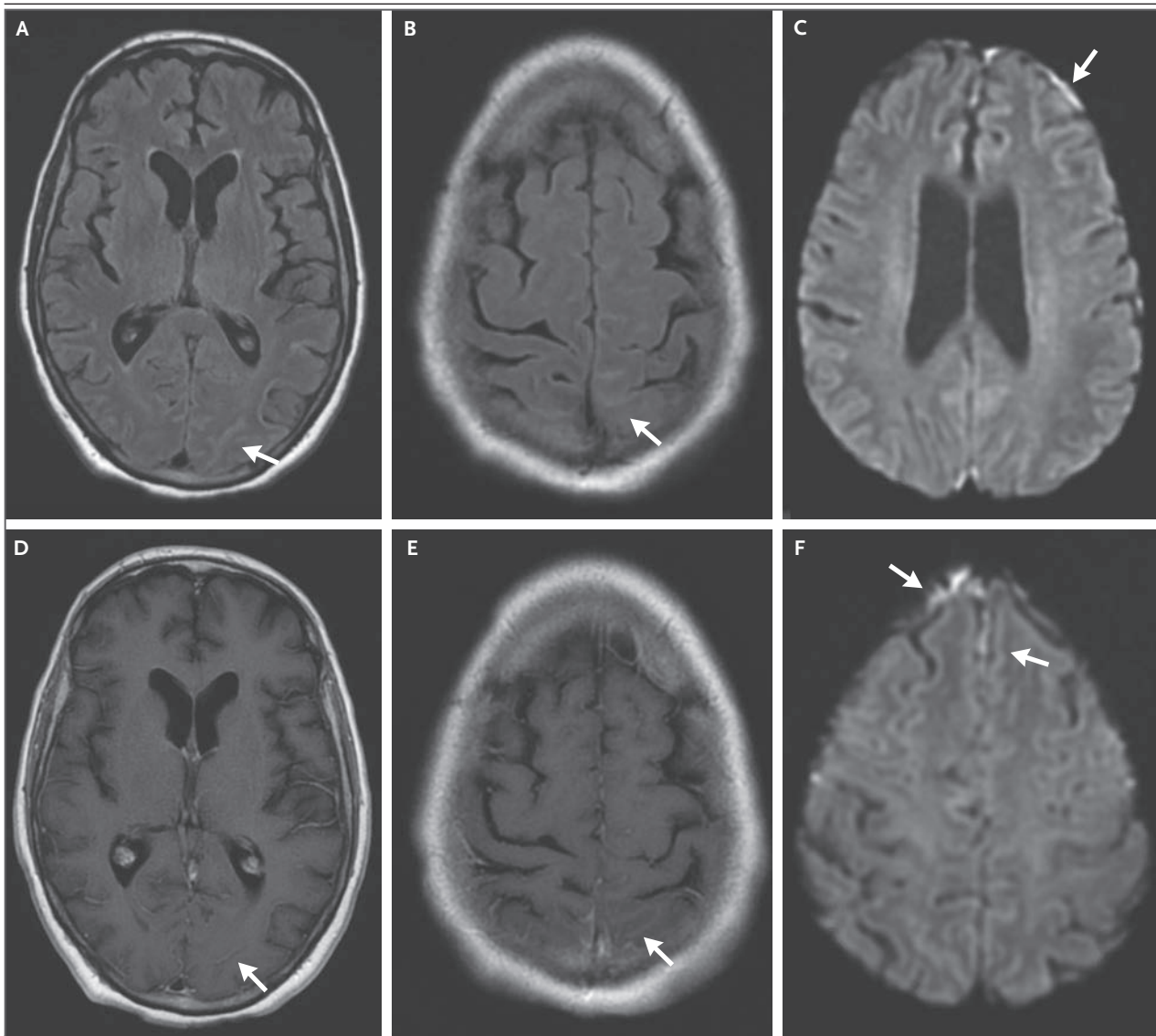


Figure 1. MRI of the Head.

Fluid-attenuated inversion recovery (FLAIR) images (Panels A and B) show hyperintensity in left occipital and parietal sulci, respectively (arrows). Gadolinium-enhanced T1-weighted images (Panels D and E) obtained at levels corresponding to those of the FLAIR images show subtle associated enhancement (arrows). Diffusion-weighted images (Panels C and F) show hyperintensity anterior to the bilateral frontal gyri and in the interhemispheric fissure that is consistent with restricted diffusion (arrows).

calm and cooperative; oriented to person, place, and time; and able to follow complex commands. CSF tests for listeria antibodies, Epstein-Barr virus DNA, and human herpesvirus 6 DNA were negative, as were blood and urine tests for histoplasma antigens. A whole-body positron-emission tomographic (PET)-CT scan did not show fluorodeoxyglucose avidity. The results of a repeat lumbar puncture are shown in Table 2.

A repeat ophthalmologic examination revealed the presence of vitreous cells, which were more numerous in the left eye than in the right eye; persistent edema in the left optic nerve; and an increase in the blind spot of the left eye on Humphrey visual-field testing. A fluorescein angiogram showed no evidence of ocular vasculitis.

A diagnosis was made.

DIFFERENTIAL DIAGNOSIS

Dr. Shibani S. Mukerji: This 63-year-old man, who had a history of primary membranous nephropathy and had received rituximab, began to have painful retro-orbital visual impairment, subacute progressive encephalopathy, and gait disturbances. Three months after the onset of symptoms, CSF evaluation was notable for lymphocytic pleocytosis without hypoglycorrhachia. His hospital course was also notable for the presence of vitreous cells in both eyes on the second ocular examination. Administration of broad-spectrum antimicrobial agents resulted in rapid improvement in his clinical condition within days despite increasing neurologic symptoms over the course of several months, findings that suggested meningoencephalitis. Despite an extensive evaluation for likely causes of meningoencephalitis, a definitive diagnosis was not established. This patient's presentation and clinical course are emblematic of challenges faced by clinicians, given that the causative agent in meningoencephalitis is identified in only 30 to 60% of cases, despite extensive and invasive testing.^{1,2}

To frame this discussion, I will focus on three salient clinical features of this patient's presentation as I consider various diagnoses in this case: uveitis associated with meningoencephalitis, subacute cognitive decline, and clinical improvement after the administration of antimicrobial therapy.

NONINFECTIOUS IMMUNE-MEDIATED CONDITIONS

Given this patient's history of autoimmune glomerular disease, it is reasonable to consider a noninfectious immune-mediated neurologic disorder as a possible diagnosis. However, the relatively prolonged course of his symptoms and the rapid improvement after the administration of antimicrobial therapy make such a diagnosis unlikely. Central nervous system (CNS) vasculitis, either primary (isolated angiitis of the CNS) or as part of systemic illness, can cause subacute dementia with headache. The clinical course of isolated angiitis of the CNS can be one of relapsing events, elevated leukocyte counts (typically lymphocytes) with elevated protein levels, and subacute clinical dementia.³ Other systemic diseases, such as eosinophilic granulomatosis with polyangiitis or granulomatosis with polyangiitis, should also be considered in this case. Tests for

ANCA and for rheumatologic disorders were negative, and the erythrocyte sedimentation rate and complement level were normal. It is also notable that this patient had received rituximab and cyclophosphamide — medications commonly used in the treatment of noninfectious vasculitic conditions affecting the CNS — yet still had clinical progression. Many such conditions are associated with angiographic changes in the intracranial blood vessels and parenchymal abnormalities that can be seen on MRI of the head. Although a small pontine infarct and sulcal hyperintensities were observed, there were no meningeal or parenchymal enhancing lesions, other subcortical ischemic infarcts, or blood vessel abnormalities that would suggest an underlying vasculitis.

CNS LYMPHOMA

The patient's progressive decline, fevers, and unintentional weight loss could suggest cancer, and extensive diagnostic testing was performed, including whole-body PET-CT. Primary CNS lymphoma is a rare form of extranodal non-Hodgkin's lymphoma that is often confined to the brain, spinal cord, and eyes, without evidence of systemic disease. In addition to headaches and behavioral changes, patients with CNS lymphoma typically have enhancing brain lesions, leptomeningeal enhancement, or other abnormalities on MRI. In contrast, patients with intravascular large B-cell lymphoma can have nondescript, insidious symptoms, without abnormalities on imaging of the head. Both types of lymphoma are treated with chemotherapy and would not result in rapid amelioration after treatment with antibiotics.

INFECTION

The most likely cause of this patient's meningoencephalitis is an infectious agent, not only because of the inflammatory nature of the CSF but also given the improvement in his clinical condition after antimicrobial therapy. Many common pathogens that are associated with uveitis and meningoencephalitis were considered, and testing for these pathogens was performed as part of this patient's evaluation. However, a unique feature of this patient's presentation is his exposure to rituximab, a humanized chimeric anti-CD20 monoclonal antibody that causes B-cell depletion. The effects of rituximab should be

considered when interpreting the results of IgG and IgM serologic tests. This concern is relevant to testing for West Nile virus infection and eastern equine encephalitis, both of which can cause neuroinvasive viral encephalitis and are endemic in the northeastern United States. The antibody response during these infections can be delayed or absent in patients with B-cell depletion.^{4,5} Such a response may also occur in Powassan virus infection, an emerging cause of viral meningoencephalitis in the United States that is transmitted by ticks.⁶ In this case, CSF antibody and RNA testing for West Nile virus and CSF testing for eastern equine encephalitis antibodies were negative. Given that West Nile virus infection, eastern equine encephalitis, and Powassan virus infection typically result in a rapidly progressive meningoencephalitis, are associated with high complication rates, and would not abate with antiviral therapy, these conditions are unlikely causes of this patient's illness.

A key question remains: What pathogen can cause uveitis and meningoencephalitis and result in rapid clinical improvement after the administration of vancomycin, ampicillin, ceftriaxone, and acyclovir? I ruled out fungi, parasites, and amoebas as possible causative agents, because they typically produce lesions that are visible on radiography, are associated with high rates of death and complications, and do not resolve without appropriate therapy.

Among the treatable infectious agents that can cause uveitis and meningoencephalitis, HSV, VZV, and CMV contribute to the worldwide burden of eye-related diseases. Infection with HSV is the most common cause of sporadic viral encephalitis, and changes in the brain that are visible on MRI are noted within days of neurologic decline. The sensitivity of HSV-specific nucleic acid testing to diagnose HSV-related meningoencephalitis is greater than 90%; the negative test in this patient makes this diagnosis unlikely. VZV infection has been associated with indolent encephalitis in immunosuppressed hosts, and a rash is not always present (zoster sine herpete). In contrast to HSV-1- and HSV-2-specific nucleic acid tests, CSF nucleic acid testing for VZV has a sensitivity of only 60% (at best) to diagnose meningoencephalitis. VZV serologic testing can aid in the diagnosis of ischemic stroke and vasculopathy; however, ocular infections due to her-

pes viruses typically result in panuveitis, and intraocular findings include necrosis of the retina and retinal periarteritis. Immunocompromised hosts may present with relatively reduced ocular inflammation, but necrotizing retinal lesions would still be expected to be present. This patient underwent multiple ophthalmologic examinations, including fluorescence angiography, with no evidence of a retinal pathologic condition, which suggests that a herpes virus is unlikely to be a cause of this patient's symptoms, and therefore treatment with acyclovir is unlikely to explain his clinical improvement.

This patient's ocular inflammation was confined to the vitreous, which suggests an intermediate uveitis, and he had optic-nerve edema. Infectious causes of intermediate uveitis include tuberculous and nontuberculous mycobacteria and Whipple's disease caused by *T. whipplei*. This patient had no epidemiologic or clinical clues — such as disturbances of ocular movement or movement disorders, or absence of granulomatous disease in the CNS or other organs — suggestive of these diagnoses. In addition, the antimicrobial agents administered in this case would not be adequate to treat a CNS mycobacterial infection.

Spirochete infections can cause uveitis and meningoencephalitis, and many organisms in this phylum, such as *T. pallidum*, leptospira, and several borrelia species, can cause pathogenic disease in humans. Syphilis can affect any portion of the eye; it can cause uveitis, which typically involves the anterior chamber, or panuveitis. Other manifestations of syphilis include chorioretinitis, optic neuropathy, retinal vasculitis, and interstitial keratitis, and if untreated, it can result in ocular hypertension and vision loss. The incidence of syphilis in the United States has increased in recent years, and delay in diagnosis has been associated with an increased risk of complications.⁷ In this patient, the *T. pallidum* antibody screen and CSF VDRL tests were negative on multiple occasions; given his immunosuppressed status, one may have anticipated more marked ocular pathologic findings if the patient had syphilis that had been untreated for 3 months.

Leptospirosis is an unlikely diagnosis. Short-term ocular findings associated with leptospirosis typically include nonspecific conjunctivitis,

and chronic phases of the disease can include anterior uveitis and neuroretinitis. In the later chronic phase involving encephalitis, changes in the brain are visible on MRI; however, no such changes are present in this patient, and there is no reported history of travel to areas in which leptospira are endemic.

In more rare instances, *B. burgdorferi* — the causative agent for Lyme disease — can cause uveitis and optic neuropathy.⁸ This patient lives in a region of the country where tickborne diseases are endemic, and he is an avid hunter. Given that his condition improved dramatically after the administration of antibiotics, specifically ceftriaxone, infection with borrelia species is a seemingly logical diagnosis for this patient. Testing for Lyme disease occurs as a part of a two-tiered algorithm and measures a person's antibody response to the spirochete. Whether treatment with rituximab delays formation of antibodies in blood and CSF is unknown, thus complicating the interpretation of this patient's serologic test results. In published cases of patients with clinically suspected *B. burgdorferi* infection who had neurologic complications consistent with Lyme disease in the context of immunosuppression, abnormal findings were observed on imaging of the head or spine. These included enhancement of the meninges, the presence of plaquelike subcortical white-matter disease, and in some patients, enhancement of cranial or spinal nerve roots.⁹ The absence of radiographic abnormalities of the CNS in this patient, despite a subacute cognitive decline with ataxia, and repeatedly negative serologic tests make *B. burgdorferi* unlikely.

B. MIYAMOTOI

There is another borrelia species that can be transmitted by the ixodes tick, and infection causes a symptom complex that is remarkably consistent with this patient's clinical presentation. In two case reports of meningoencephalitis in immunocompromised patients receiving rituximab, *B. miyamotoi* was the causative agent.^{10,11} The patients described in the published reports were 70 and 80 years of age, had received rituximab for hematologic cancers, and presented with weight loss, subacute cognitive decline, gait disturbances, lymphocytic pleocytosis, and an elevated total protein level in the CSF. Both patients recovered after the administration of in-

travenous antibiotics. A detailed description of MRI of the head was not published for either patient, but the findings of both were reportedly normal. In both patients, Wright-Giemsa staining of CSF showed spirochetes, and a definitive diagnosis of *B. miyamotoi* infection was made on the basis of nucleic acid testing of the blood.

Additional features of this patient's presentation that support a diagnosis of *B. miyamotoi* infection include recurrent fever and the lack of an associated rash. The features that do not fit well with this diagnosis are the patient's vision symptoms and ophthalmologic findings. To my knowledge, there are no reports of *B. miyamotoi* causing eye disease, but given our limited understanding of this infection and the fact that other spirochetes can cause uveitis and edema in the optic-nerve head, I suspect that *B. miyamotoi* is the most likely diagnosis in this patient. To establish this diagnosis, I would perform nucleic acid testing of the CSF for *B. miyamotoi* DNA while continuing treatment with intravenous ceftriaxone.

DR. SHIBANI S. MUKERJI'S DIAGNOSIS

Borrelia miyamotoi infection.

PATHOLOGICAL DISCUSSION

Dr. John A. Branda: A lumbar puncture was performed on hospital day 2 for CSF analysis. A real-time polymerase-chain-reaction assay targeting the *glpQ* (glycerophosphodiester phosphodiesterase) gene of borreliae that cause relapsing fever — a gene that is absent in Lyme disease-related borreliae¹² — was positive. According to a melting-curve analysis, the detected *glpQ* DNA was derived from *B. miyamotoi* rather than other borrelia species in the relapsing-fever group. Serum obtained on hospital day 2 also showed strong reactivity on an enzyme-linked immunosorbent assay (ELISA) that detects IgG antibodies directed against the GlpQ protein of *B. miyamotoi*. The corresponding IgM ELISA was negative, consistent with *B. miyamotoi* infection of several months' duration.^{13,14}

MICROBIOLOGIC DIAGNOSIS

Borrelia miyamotoi infection.

MANAGEMENT AND FOLLOW-UP

Dr. Ard: B. miyamotoi, a spirochete that was first identified in 1995 in Japan,¹⁵ has emerged as a cause of human disease on multiple continents. In 2011, the organism was first reported as the cause of human illness in 46 patients with fever in Russia,¹⁶ and in 2013, the first case of *B. miyamotoi* was identified in the United States¹⁰; the similarities between the 2013 case and the current one are striking. Both patients had received rituximab and presented with subacute meningoencephalitis and a lymphocytic pleocytosis in the CSF that abated with antimicrobial therapy.

I am not aware of any randomized, controlled trials of treatment for *B. miyamotoi*, and there are no formal treatment recommendations. Patients with *B. miyamotoi* infection typically receive doxycycline, ceftriaxone, or amoxicillin — treatments commonly used for Lyme disease. One study showed that *B. miyamotoi* was susceptible in vitro to doxycycline, azithromycin, and ceftriaxone but

not to amoxicillin¹⁷; the clinical significance of these findings has not been established.

This patient was discharged home with instructions to complete 4 weeks of treatment with intravenous ceftriaxone. However, a facial rash developed, and treatment was changed to doxycycline. At a follow-up appointment 3 weeks after hospital discharge, all the patient's symptoms except for blurry vision had resolved, and his neurologic examination was normal. His vision improved slowly over the course of 3 months.

FINAL DIAGNOSIS

Borrelia miyamotoi infection.

This case was presented at an Internal Medicine Continuing Medical Education course.

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